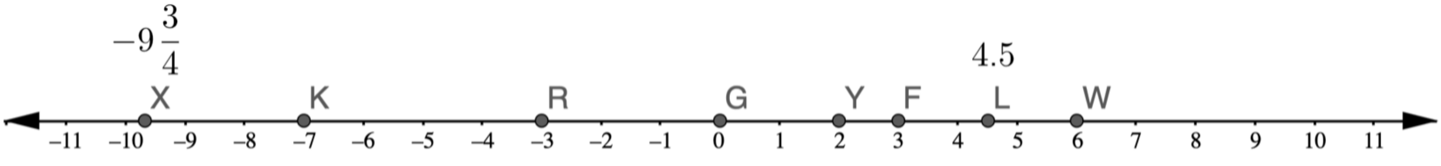
Grade 6 Accelerated | Name____________________
Unit 3 | Date____________________
Lesson 1 Practice | Period____________________

A number line with points X, K, R, G, Y, F, L, and W is shown.



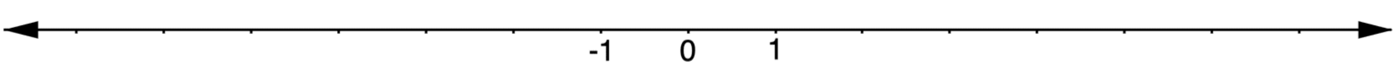
1. Complete the table by identifying which points belong in each category. Some points may be used more than once and some may not be used at all.

Natural Numbers	Whole Numbers	Integers	Rational Numbers

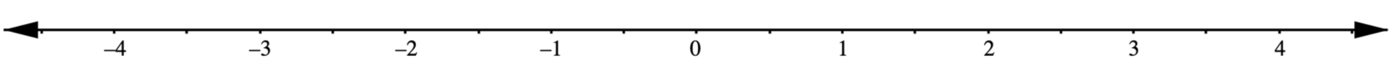
2. Complete the table by identifying which points belong in each category. Some points may be used more than once and some may not be used at all.

Positive Numbers	Negative Numbers

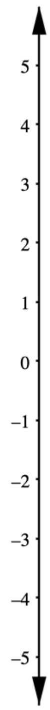
3. Plot the integers $6, -4, -2,$ and -5 on the number line.



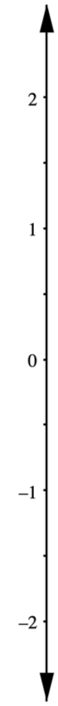
4. Plot the numbers $\frac{2}{3}$, -2.25, 3.6, and $-2\frac{1}{2}$ on the number line.



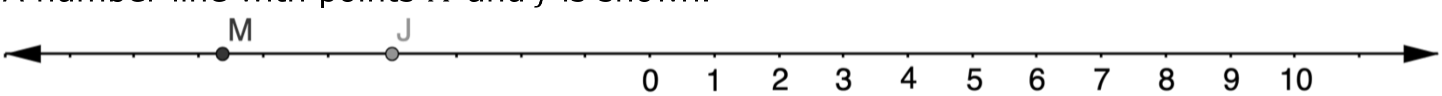
5. Plot the numbers $-\frac{5}{2}$, $4\frac{2}{3}$, and -1.25



6. Plot the numbers $\frac{7}{4}$, $-\frac{7}{8}$, and -1.2



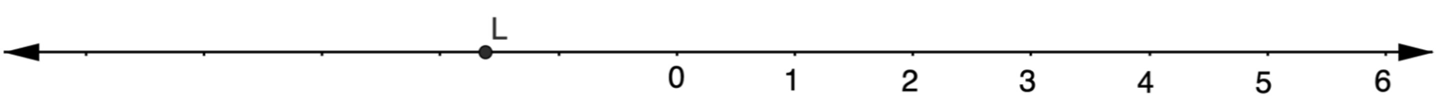
A number line with points M and J is shown.



7. Explain how to estimate the value of J.

8. Explain how to estimate the value of M.

Jimmy and Luna both estimated the value of L on the number line shown.



9. Jimmy says the value of L is about $-2\frac{1}{3}$. Explain whether you agree with Jimmy.

10. Luna says the value of L is about $1\frac{2}{3}$. Explain whether you agree with Luna.

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Lesson 2 Practice

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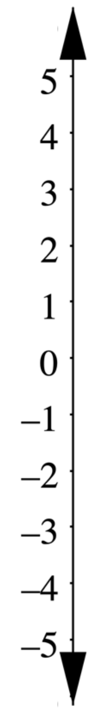
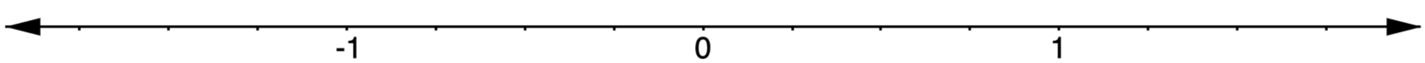
Use the vertical number line to answer the following problems.



1. Plot 3.25 and its opposite.

2. Which is larger, 3.25 or its opposite?

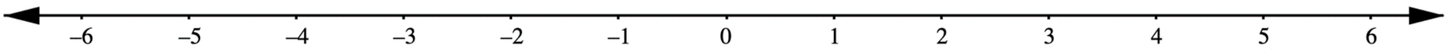
Use the horizontal number line to answer the following problems.



3. Plot $-1\frac{3}{8}$ and its opposite.

4. Which is larger, $-1\frac{3}{8}$ or its opposite?

5. Plot and label each value and its opposite on the number line: -1.6, $3\frac{2}{3}$, 0.5, $-\frac{5}{2}$



Complete each comparison statement for the pairs of numbers you plotted in question 5. Use $<$ or $>$ in the first blank and the given number's opposite in the second blank.

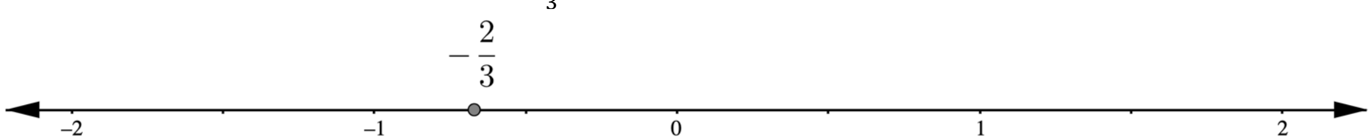
6. -1.6_____ ________

7. $3\frac{2}{3}$ _____ ________

8. 0.5 _____ ________

9. $-\frac{5}{2}$ _____ ________

10. A number line with the value $-\frac{2}{3}$ plotted is shown.



Stefan said that the opposite of $-\frac{2}{3}$ is $\frac{1}{3}$ because it's on the other side of 0. Explain whether you agree with Stefan.

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Lesson 3 Practice

Name____________________
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Explain why each inequality statement is true or false.

1. $-0.45 > -1.2$

2. $72\% < -8\%$

Complete each statement with an inequality symbol.

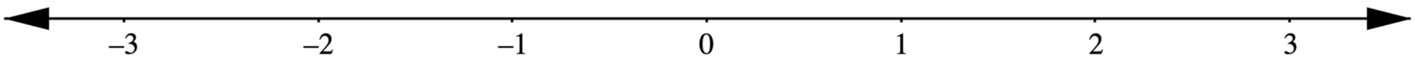
3. 12.3 _____ 125%

4. -9.45 _____ -9.88

A set of rational numbers is shown.

$$-1\frac{2}{3}, \quad -1.8, \quad \frac{7}{8}, \quad -\frac{3}{5}, \quad 0.75$$

5. Plot each of the values on the number line. Label each point with its numeric value.



6. Complete the inequality statement to show the relationship between the values.

$$\underline{\hspace{2cm}} < \underline{\hspace{2cm}} < \underline{\hspace{2cm}} < \underline{\hspace{2cm}}$$

Complete each statement using <, >, or =.

7. $-\frac{7}{5}$ ______ $-1\frac{2}{5}$

8. -1.4 ______ $-\frac{3}{2}$

9. Order the values from least to greatest.

$$-2\frac{2}{3},\quad \frac{11}{4},\quad -\frac{9}{2},\quad -\frac{3}{5},\quad 0,\quad 2\frac{1}{3}$$

10. Order the values from least to greatest.

$$-3.75,\quad \frac{5}{3},\quad -3\frac{4}{5},\quad -2.6,\quad 2\frac{3}{5}$$

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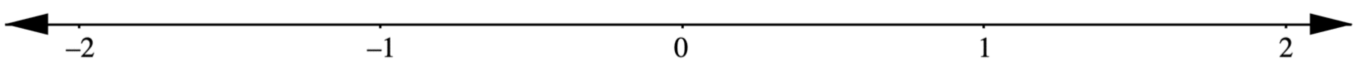
A set of rational numbers is shown.

$$-\frac{6}{5}, \quad -1.3, \quad 70\%, \quad -1\frac{1}{3}, \quad \frac{5}{8}, \quad -150\%, \quad 0.9$$

1. Plot each of the values on the number line. Label each point with its numeric value.



2. Order the values from least to greatest.

Complete each statement using $<$, $>$, or $=$.

3. -8.88 ______ $-8\frac{9}{10}$

4. 2.4% ______ $\frac{1}{5}$

5. -67% ______ 0.67

6. -52% ______ $-\frac{13}{25}$

7. -3.472 ______ $-3\frac{5}{12}$

8. -12% ______ $-1\frac{1}{5}$

9. Order the values from least to greatest.

$$-\frac{5}{9}, \quad 65\%, \quad -\frac{3}{10}, \quad -0.451, \quad 0.815, \quad \frac{3}{5}$$

10. Luka wrote the inequality statement:

$$-4.111 < -450\% < -\frac{14}{3}$$

Explain whether you agree with Luka.

Grade 6 Accelerated
Unit 3
Lesson 5 Practice

Name____________________
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Write a value that represents each description.

1. A mountain climber standing $12,840$ feet above sea level

2. A temperature decrease of $3.5°$

3. A deposit of $\$372.65$ to a savings account

4. An expense of $\$49.33$

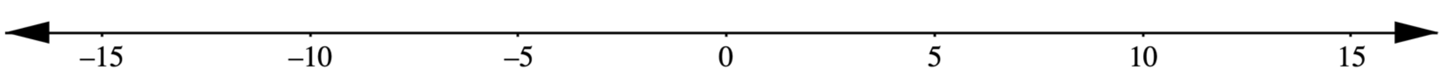
In the last two plays of a football game, the offense had a gain of 7 yards and then a loss of 12 yards.

5. Write a rational number that represents the football's change in position with each play.

   Gain of 7 yards: ____________

   Loss of 12 yards: _____________

6. Plot both rational numbers on the number line.



7. What does 0 represent in this context?

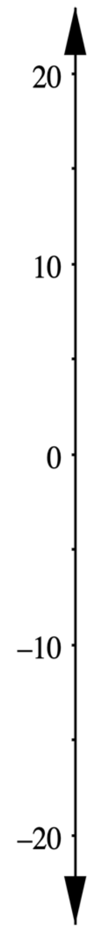
A boat is on the water in Crystal Keys. The top of the mast of the boat is 15 feet above sea level, and the bottom of the boat is 6 feet below sea level.

8. Write a rational number that represents the elevation of each part of the boat.

Top of the mast: ____________

Bottom of the boat: _____________

9. Plot both rational numbers on the number line.



10. What does 0 represent in this context?

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Unit 3
Lesson 6 Practice

Name____________________
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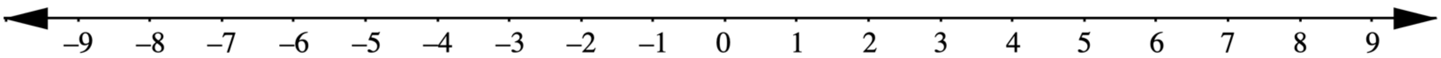
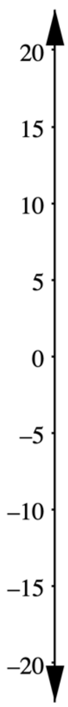
1. Plot 7 and its opposite on the number line.

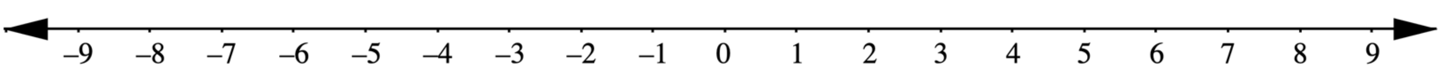


2. On the number line, plot a location of 14 feet above sea level and its opposite.



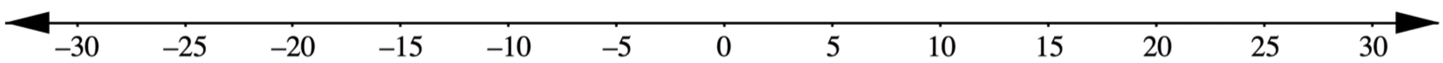
What is the meaning of 0 in this context?

3. Plot a decrease of $4.75°$ and its opposite on the number line. What is the meaning of 0 in this context?



From City Hall, the driving range is 23 blocks East and the train station is 23 blocks West.

4. Model all three locations on the number line.



5. What does 0 represent in this situation?

6. From City Hall, is it farther to the driving range or to the train station?

Complete the table for each description.

	Description	**Rational Number**	**Opposite**	**Meaning of Opposite**	**Meaning of 0**
7.	A withdrawal of $\$94.72$				
8.	A gain of 1.2 pounds				
9.	397 yards above sea level				
10.	A temperature decrease of $9.2°$				

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Lesson 7 Practice

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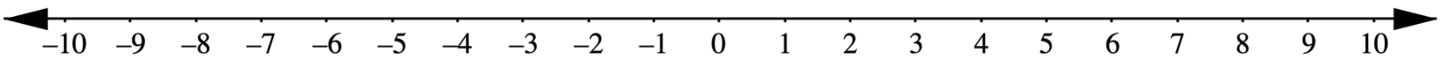
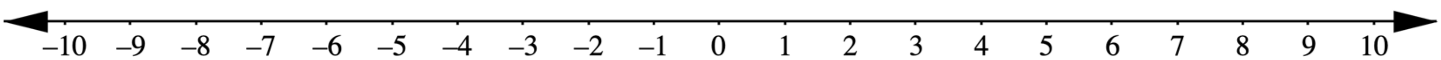
Complete each statement and model it on the number line.

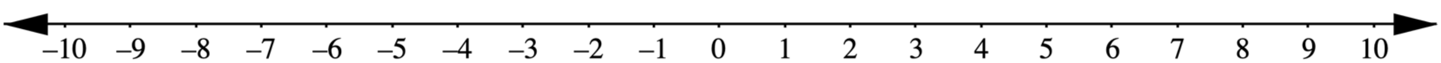
1. $|9| =$ ______



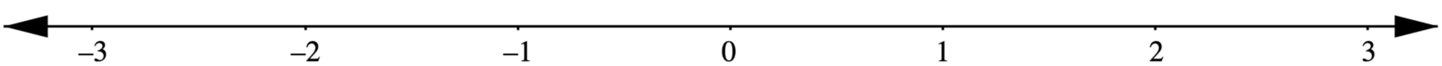
2. $|-3.5| =$ ______



3. $|8.25| =$ ______



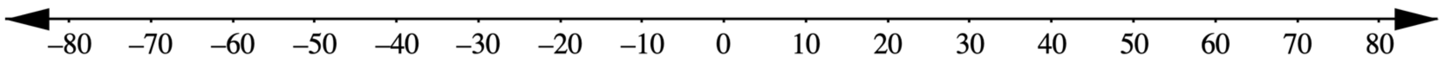
4. $\left|\frac{7}{3}\right| =$ ______



5. Martina claims that an absolute value cannot be negative. Explain whether you agree with Martina.

Raechelle makes a withdrawal of $\$59.74$ from her checking account.

6. Model Raechelle's transaction on the number line.



7. What does the absolute value of Raechelle's transaction mean in this context?

8. From noon to midnight, the temperature change in Lakeland, Fl was $-8.5°\mathrm{F}$. What does the value $|-8.5|°\mathrm{F}$ mean in this context?

9. Describe the meaning of each value in the context of elevation.

Value	**Interpretation**
$5{,}362$ feet	
-743 feet	
$\|67\|$ feet	
$\|-3{,}918\|$ feet	

10. The absolute value of Amarya's account balance is $\$75.88$. What could be Amarya's account balance?

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Use the symbols <, >, and = to compare each pair of values.

1. $|-83|$ _____ -77

2. 31.8 ______ $|-31.8|$

Find the value of each expression.

3. $|-44| - |-18|$

4. $|-5| \times |-9|$

5. $|-23| + |-7| + 2$

Sia made two transactions on her bank account last week. She made a withdrawal of $\$184.60$ and a deposit of $\$98.55$.

6. Write a value that could be used to describe each of Sia's transactions.

Transaction	**Value**
Withdrawal	
Deposit	

7. Use an inequality symbol, $>$ or $<$, to compare the absolute values of Sia's transactions.

|_____| _____ |_____|

8. What does this inequality statement indicate about the total change in Sia's bank account last week?

On April 6th, 2020, two baby giraffes were born at the Miami zoo to two different female giraffes, Mia and Zuri. Mia's calf was 49 pounds above the average newborn giraffe weight, while Zuri's calf was 13 pounds below the average weight.

9. Write a value that could be used to describe each calf's weight in relation to the average.

Calf	**Value**
Mia's calf	
Zuri's calf	

10. Use an inequality symbol, $>$ or $<$, to compare the absolute values.

|_____| _____ |_____|

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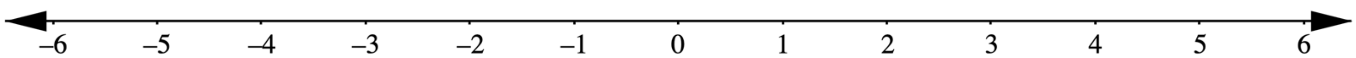
The coffee shop is 5 blocks East of Amber’s house. The park is 3 blocks West of Amber’s house.

1. Model Amber’s house, the coffee shop, and the park on the number line.



2. Write a mathematical statement using absolute value that could be used to find the distance from the coffee shop to the park.

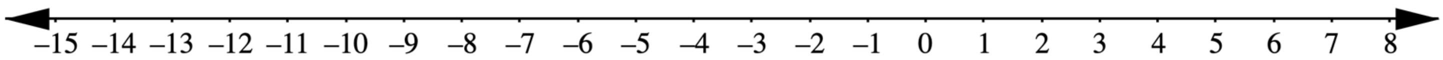
3. How many blocks is it from the coffee shop to the park?

4. Write a mathematical statement using absolute value that could be used to determine how much farther from Amber’s house the coffee shop is than the park.

5. How much farther from Amber’s house is the coffee shop than the park?

For the 4 hours between 5pm and 9pm, the temperature change was $-2°$ per hour.

6. Model the change in temperature on the number line.



7. Write a mathematical statement using absolute value that could be used to determine the total decrease in temperature from 5pm to 9pm.

In 2020, the country with the greatest increase in population by percent was Niger, with an increase of 3.84%. The country with the greatest decrease was Lithuania, with a decrease of 1.35%.

8. Write a value that could be used to describe each country's net change in population by percent.

Country	**Net Change in Population by Percent**
Niger	
Lithuania	

9. Use an inequality symbol, $>$ or $<$, to compare the absolute values of the population changes by percent for each country.

|_____| _____ |_____|

10. Which country's population changed more by percentage?